Does “not good” confuse a model?

A simple NLP experiment

Positive and negative restaurant reviews

Compare training data before and after adding examples

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Our question

Can extra examples with “not” help?

“The food is good.” = positive

“The food is not good.” = negative

We add examples and compare the results.

The program predicts positive or negative.

Our data

Short English restaurant reviews from UCI

- 1,000 reviews in the original Yelp file
- 996 different reviews after removing duplicates
- 796 reviews to train / 200 reviews to test
- Both models use the same test reviews.

How we made two models

The same algorithm, with different training data

First model: 796 original training reviews

Second model: 796 reviews plus 120 new examples

Example: “The food was not good.” = negative

Example: “The food was not bad.” = positive

A simple model learns from words and word pairs.

Results

Accuracy = percentage of correct answers

Real reviews: 84.5% before / 81.0% after

Real reviews with negation words: 75.6% / 75.6%

Artificial examples: 41.7% before / 93.3% after

Artificial examples reuse the same important words.

They do not prove the model understands new phrases.

How the program works

A short Python program

Read reviews and split them into training and test data.

Create extra examples with “not”.

Train and test both models.

Enter an English review to compare their predictions.

Demo

Compare the two models on our website

Open: psikonday.github.io/isnusl/

Enter: The food was not bad.

Click Analyze review to see both predictions.

The results show sentiment before and after augmentation.

The models use parameters learned earlier in Python.

What did we learn?

The result has an important limit

The artificial test score increased.

The score on all real reviews decreased.

We need more varied examples and more testing.

Source: UCI Sentiment Labelled Sentences

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